

BACCHIRI!—Verifiable Measurement Layer

Show threshold compliance. Keep sensor values private.

A third party can check the day, public threshold, and result. Sensor values remain hidden.

SENSOR VALUES

STAY ON EDGE DEVICE

THIRD PARTY CHECKS

DAY · THRESHOLD · RESULT

WAVE 1

**MIDNIGHT PREPRODUCTION
NETWORK VERIFIED**

Wave 1 · Source validated 2026-08-29 JST · Midnight record 2026-08-28 JST

PROVE THRESHOLD COMPLIANCE WITHOUT DISCLOSING SENSOR VALUES

CONVENTIONAL: SHARE RAW DATA



Raw sensor values

Share with
third party

Unnecessary
disclosure

VERIFIABLE SENSOR PROOF

EDGE DEVICE



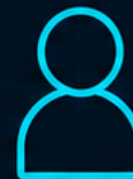
Sensor values
stay on device

ZERO-KNOWLEDGE PROOF
(VERIFY WITHOUT REVEALING)



Checks registered
threshold

THIRD PARTY



Checks day ·
threshold · result

✓ WITHIN

! OUTSIDE

◻ STOPPED



THIRD PARTY SEES: DAY · THRESHOLD · RESULT

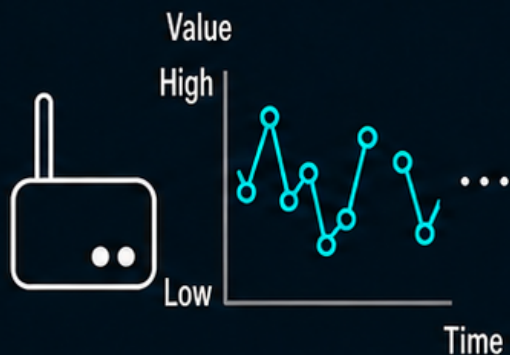


THIRD PARTY DOES NOT SEE: SENSOR VALUES

HOW THE ZERO-KNOWLEDGE PROOF IS MADE

1 SENSOR VALUES

Many readings stay on the Edge Device



2 GROUP BY HOUR

Hourly minimum · maximum · count
Arrange one day into 24 hourly slots



3 CHECK WITHOUT REVEALING

Hourly minimum · maximum
Compare with registered threshold



4 PUBLISH ONLY NEEDED INFORMATION

Publish day · threshold · result
Do not publish sensor values



PROOF: Compare each observed hour's minimum / maximum with the registered threshold and determine WITHIN / OUTSIDE



DOES NOT PROVE: Sensor accuracy · Measurement completeness · Aggregation correctness

WHAT THE ZERO-KNOWLEDGE PROOF DOES — AND DOES NOT — SHOW

PROVES



Decision for each observed hour



Check against registered threshold



WITHIN • OUTSIDE



Day • Device • Threshold

STAYS PRIVATE



Raw sensor values



Hourly minimum • maximum



Nonce • private proof data



Time and value causing OUTSIDE

DOES NOT PROVE



Physical sensor integrity



Calibration • installation quality



Continuous operation • measurement completeness



Edge aggregation correctness



EXACT CLAIM: Compare each observed hour's minimum / maximum with the registered threshold and determine WITHIN / OUTSIDE

WAVE 1 COMPONENTS AND RESPONSIBILITIES



EDGE DEVICE

Measurement · Hourly aggregation ·
Transaction signing



FRONTEND

Administrator actions ·
Third-party view



BACKEND

Public-record storage ·
Proof generation



MIDNIGHT

Threshold · Target Device ·
Result record

This page shows responsibilities only. Disclosure is on the next page; the processing sequence is on page 7.

DATA × VIEWER: STORAGE AND DISCLOSURE

DATA	STORAGE	ADMINISTRATOR	THIRD PARTY	MIDNIGHT RECORD
Raw sensor values	Edge Device	PRIVATE	PRIVATE	NOT RECORDED
Hourly minimum · maximum	Edge Device / Backend during proof	AUTHENTICATED ONLY	PRIVATE	NOT RECORDED
Proof metadata (Day · Observed hours · Count · Spec version)	Midnight / D1 (display mirror)	PUBLIC	PUBLIC	PUBLIC RECORD
Public threshold (Lower · Upper · Unit)	Midnight / D1 (display mirror)	PUBLIC	PUBLIC	PUBLIC RECORD
Result	Midnight / D1 (display mirror)	PUBLIC	PUBLIC	PUBLIC RECORD
Signed transaction	Midnight / D1 (display mirror)	PUBLIC	PUBLIC	PUBLIC RECORD

 **PUBLIC:** Anyone can check

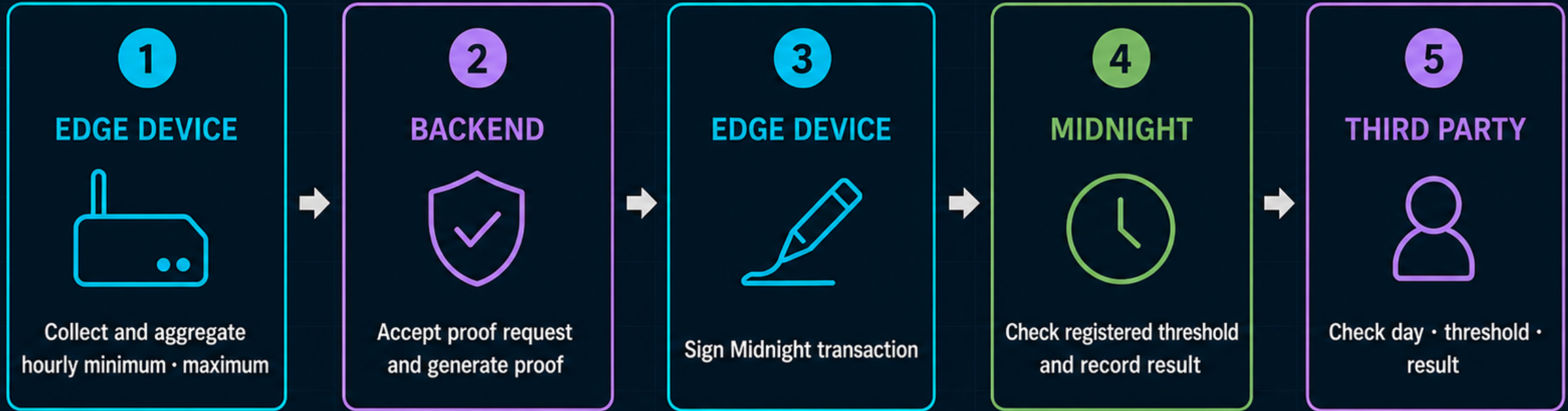
 **AUTHENTICATED ONLY:** Administrator only

 **PRIVATE:** Not shown to third parties

D1 is a display mirror. Third parties rely on the public Midnight record.

Backend handles hourly minima / maxima only during proof generation and does not store them.

DAILY PROOF FLOW



PUBLIC: Day · Threshold · Result ·
Midnight transaction record



PRIVATE: Hourly minimum ·
maximum are not shown to third parties

Show the final GUI from action to public verification

Capture follows the completed GUI; required review evidence is fixed now.

FINAL IMPLEMENTED GUI CAPTURE

Insert after the GUI freeze

Device operation · Proof request · Midnight transaction · Third-party review

VIDEO PRODUCTION IS LAST

FOUR REQUIRED CAPTURES

- 01 Device-to-threshold binding
- 02 One day proved in 24 hourly slots
- 03 Device-signed Midnight transaction
- 04 Public view hides MIN / MAX values

Dashboard tests: 26 / 26 PASS

VALIDATED IMPLEMENTATION

1

6 PROOF CIRCUITS

✓ **ALL COMPILE**

Includes the daily threshold proof

2

128 AUTOMATED TESTS

✓ **ALL PASS**

Across 8 workspaces

3

TYPECHECK + BUILD

✓ **ALL PASS**

Cloudflare pre-deployment check also passes

4

MIDNIGHT PREPRODUCTION NETWORK

✓ **VERIFIED 2026-08-28**

WITHIN · OUTSIDE · Device-signed transaction



CURRENT SOURCE VALIDATION: 2026-08-29 JST



MIDNIGHT RECORD: 2026-08-28 (NOT RERUN TODAY)

THREE-STAGE DELIVERY PLAN

CURRENT

WAVE 1 • VERIFIED

- ✓ Device-authenticated daily proof
- ✓ Aggregate one day into 24 hourly slots
- ✓ WITHIN / OUTSIDE confirmed on Midnight preproduction network
- ✓ Administrator + third-party views
- ✓ Daily threshold-proof submission

PLANNED

WAVE 2 • PLANNED

- ✓ Independent browser verification
- ✓ Signed provenance proof
- ✓ Operational automation
- ✓ Recovery + multi-device monitoring

WAVE 3 • PLANNED

- ✓ Calibrated-device proof
- ✓ Calibration record + firmware identity
- ✓ Secure hardware
- ✓ Cross-organization audit



TECHNICAL VALIDATION

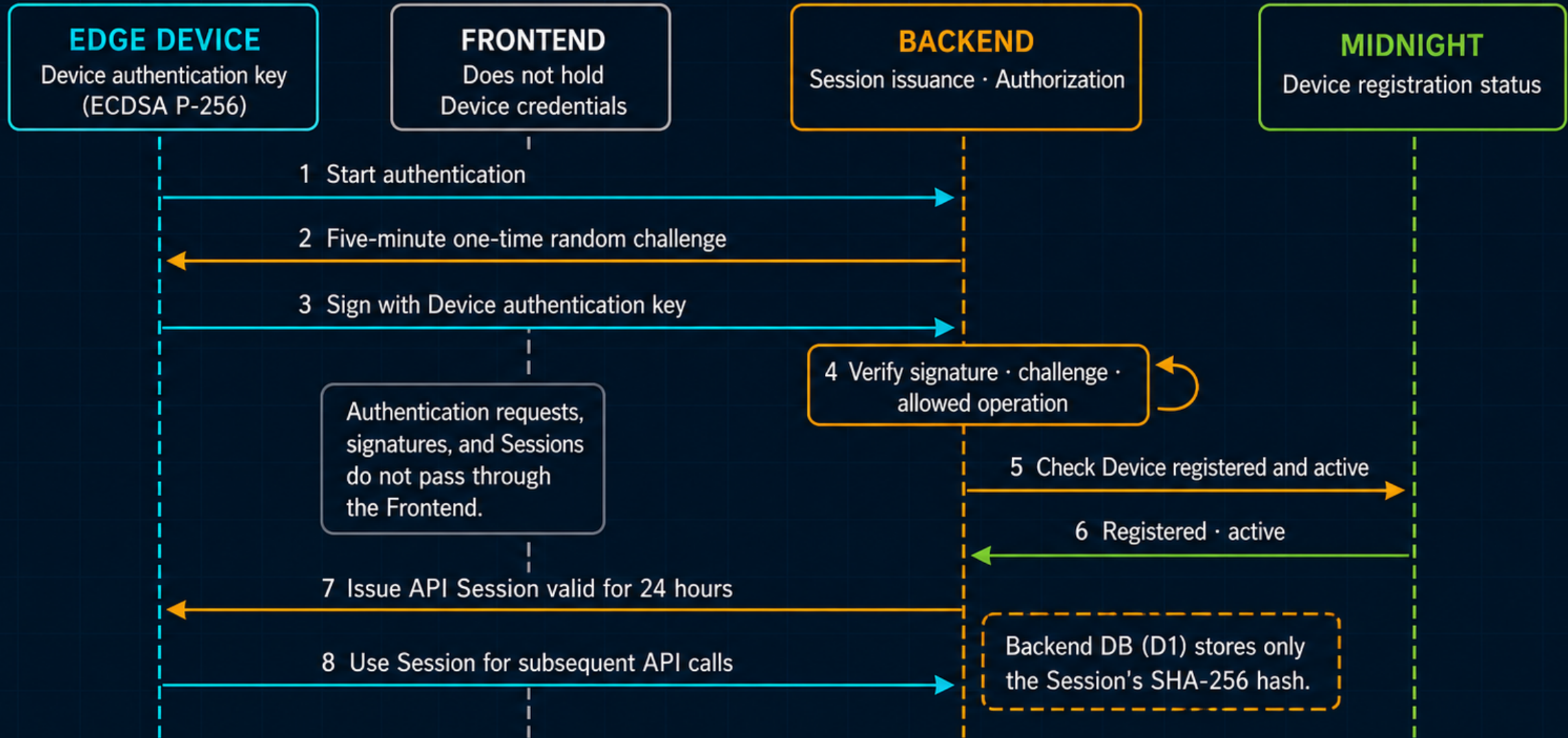


CONSTRUCTION-SITE PILOT



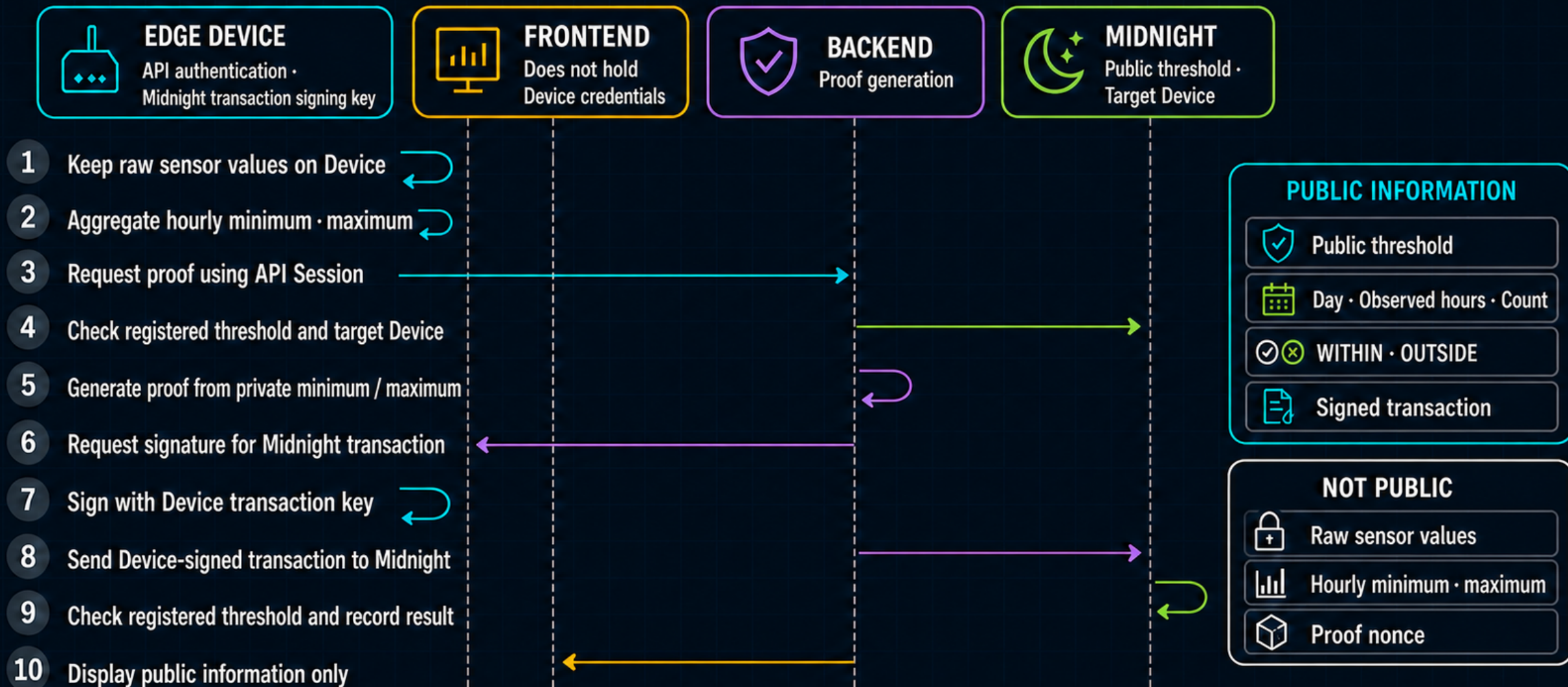
SALES + RENTAL ADOPTION

USE CASE 1: FROM DEVICE AUTHENTICATION TO API USE



Authentication key is used only to issue / renew a Session. API Session is 24-hour authentication data, not an encryption key.

USE CASE 2: FROM PRIVATE SENSOR VALUES TO PUBLIC RESULT



CUSTOMER VALUE: A third party can check day · threshold · result. Sensor values remain private.